

Overweight, obesity, and mortality from cancer in a prospectively studied cohort of U.S. adults.



BACKGROUND: The influence of excess body weight on the risk of death from cancer has not been fully characterized. **METHODS:** In a prospectively studied population of more than 900,000 U.S. adults (404,576 men and 495,477 women) who were free of cancer at enrollment in 1982, there were 57,145 deaths from cancer during 16 years of follow-up. We examined the relation in men and women between the body-mass index in 1982 and the risk of death from all cancers and from cancers at individual sites, while controlling for other risk factors in multivariate proportional-hazards models. We calculated the proportion of all deaths from cancer that was attributable to overweight and obesity in the U.S. population on the basis of risk estimates from the current study and national estimates of the prevalence of overweight and obesity in the U.S. adult population. **RESULTS:** The heaviest members of this cohort (those with a body-mass index [the weight in kilograms divided by the square of the height in meters] of at least 40) had death rates from all cancers combined that were 52 percent higher (for men) and 62 percent higher (for women) than the rates in men and women of normal weight. For men, the relative risk of death was 1.52 (95 percent confidence interval, 1.13 to 2.05); for women, the relative risk was 1.62 (95 percent confidence interval, 1.40 to 1.87). In both men and women, body-mass index was also significantly associated with higher rates of death due to cancer of the esophagus, colon and rectum, liver, gallbladder, pancreas, and kidney; the same was true for death due to non-Hodgkin's lymphoma and multiple myeloma. Significant trends of increasing risk with higher body-mass-index values were observed for death from cancers of the stomach and prostate in men and for death from cancers of the breast, uterus, cervix, and ovary in women. On the basis of associations observed in this study, we estimate that current patterns of overweight and obesity in the United States could account for 14 percent of all deaths from cancer in men and 20 percent of those in women. **CONCLUSIONS:** Increased body weight was associated with increased death rates for all cancers combined and for cancers at multiple specific sites. Copyright 2003 Massachusetts Medical Society